C# 9

↳ C.I when 'σ' known

$$\bar{X} \pm z \frac{\sigma}{\sqrt{n}}$$

↳ C.I when 'σ' unknown

$$\bar{X} \pm t \frac{s}{\sqrt{n}}$$

↳ Find sample size "n":

$$n = \left(\frac{z \times \sigma}{\varepsilon} \right)^2$$

C# 10

↳ when 'σ' known

$$z = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}}$$

↳ when 'σ' unknown:

$$t = \frac{\bar{X} - \mu}{s / \sqrt{n}}, \quad df = n-1$$

↳ P values:

- if alternat hyp. has $>$: check the value in the body of z table against calcuted z, then minus this value from 1. This is final p-value.

- if alternate hyp. has <: then check the value in the z table against calculated z ⇒ This is your final p-value.

- if alternate hyp. has ≠ : then check the value in z table body against calculated z, ⇒ minus this from 1 ⇒ then multiply this with 2.

CH 11 (Independent tests).

↳ when 'σ' known:

$$z = \frac{\overline{X_1} - \overline{X_2}}{\sqrt{\left(\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right)}}$$

↳ when 'σ' unknown:

- Case 1 (Equal Variance):

$$t = \frac{\overline{X_1} - \overline{X_2}}{\sqrt{Sp^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

$$Sp^2 = \frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{n_1+n_2-2}$$

$$df = n_1 + n_2 - 2$$

• Case 2: (Unequal variance):

$$t = \frac{\overline{X}_1 - \overline{X}_2}{\sqrt{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)}}$$

$$df = \frac{\left[(s_1^2/n_1) + (s_2^2/n_2)\right]^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}}$$

(Dependent a paired t-test)

$$t = \frac{\overline{d}}{s_d/\sqrt{n}}$$

$$df = n - 1$$

— (Extra working Formulas) —

$$s = \sqrt{\frac{\Sigma(X_i - \overline{X})}{n - 1}}$$

$$s^2 = s^2$$

C# 12

ANOVA

$G.M = \bar{X} = \dfrac{\text{Sum of all value from every group}}{\text{total No. of obs including every group}}$

For equal sample sizes:

SST or $BSS = n\left[(\bar{X}_1 - G.M)^2 + \ldots + (\bar{X}_n - G.M)^2\right]$

$n \downarrow$ size of each group

For unequal sample size:

SST or $BSS = n_1(\bar{X}_1 - G.M)^2 + \ldots\, n_n(\bar{X}_n - G.M)^2$

where n_1 = size of group 1 in unequal case.

Now Next Steps same for both equal / unequal sizes:

~~S.S.E or WSS:~~

$\cancel{\to (X_1 - \bar{X}_1)^2 + \ldots (X_n - \bar{X}_1)^2}$

SSE or WSS:

Let say 3 groups, with their means $\overline{X}_1$, $\overline{X}_2$, $\overline{X}_3$

then
take 1st group:

$$\Sigma(\text{individual groupvalue} - \overline{X}_1)^2$$

take 2nd group

$$\Sigma(\text{indv. groupvalue} - \overline{X}_2)^2$$

take 3rd group:

$$\Sigma(\text{indv. groupvalue} - \overline{X}_3)^2$$

Now
SSE or WSS = Sum of all three

ANOVA Table

Source of var.	SS	d.f	M.S	F
Treatment	SST	$n-1$	SST/df	$\frac{MST}{MSE}$
Error	SSE	$k-n$	SSE/df	

when n = No. of groups
k = total No. of obs.

C# 13

$$r = \frac{n\sum XY - \sum X \cdot \sum Y}{\sqrt{[n\sum X^2 - (\sum X)^2][n\sum Y^2 - (\sum Y)^2]}}$$

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$$

regression eq:

$$\hat{Y} = a + bX$$

$$b = \frac{n\sum XY - \sum X \cdot \sum Y}{n\sum X^2 - (\sum X)^2}$$

$$a = \overline{Y} - b\overline{X}$$

$$R^2 = 1 - \frac{\sum (Y - \hat{Y})^2}{\sum Y^2 - \frac{(\sum Y)^2}{n}}$$

$$S_e = \sqrt{\frac{\sum (Y - \hat{Y})^2}{n-2}}$$

range of r (-1 -> 1)

$r < 0.5 \rightarrow$ weak corelation

$r = 0.5 \rightarrow 0.69$ Moderate

$r = 0.7 \rightarrow 1 \rightarrow$ strong

Range of R^2

0 → 1

Acc. to AI

1

>0.90 → Perfect fit

0.70 — 0.89 → Good "

0.50 – 0.69 → Moderate "

<0.50 → Weak "

0 → Explain Nothing.